

Climate Change Trend Analysis and Forecasting

Intern Project – Scope and Weekly Requirements

IDEAS TIH Summer Internship 2026 · Mentor Reference Document · June 2026 · v4

1. Project Overview

Goal

Build an end-to-end analytical project that ingests open Greenhouse Gas (GHG) emissions datasets, performs exploratory data analysis, trains forecasting models to project future emissions, and assembles findings into a well-documented Jupyter Notebook and an optional interactive Streamlit dashboard.

Note on scope: This project deliberately focuses on classical machine learning and time-series methods — specifically regression models and Holt's Damped Trend (ETS) forecasting. These approaches are well-suited to structured tabular data and annual time-series with a limited number of data points, and provide a solid foundation before exploring more complex techniques such as deep learning or LLMs.

Core Deliverables

- **Primary:** Jupyter Notebook (fully documented with markdown cells)
- **Stretch Goal:** Streamlit interactive dashboard app — attempt only if Weeks 3–4 are fully complete and time permits (see Week 6)
- **Required:** Final Presentation to mentor (1 hour)

Datasets

- **Primary — OWID CO₂ and GHG dataset:** Annual CO₂ and GHG emissions by country (total, per capita, per GDP, consumption-based vs production-based), plus energy mix and related indicators. Source: github.com/owid/co2-data
- **Supporting — Climate Watch Historical Emissions:** climatewatchdata.org

Countries of Focus

The following 10 countries will be used consistently throughout the project for all analysis, modelling, and scenario work:

China · USA · India · Russia · Japan · Germany · Brazil · United Kingdom · South Africa · Australia

These represent a mix of major emitters, economies at different stages of development, and countries with documented emissions reduction trajectories (e.g. UK, Germany), providing a varied and instructive analytical set.

Tools and Libraries

Tool / Library	Purpose
Python 3.x, Jupyter Notebook	Primary development environment
Pandas, NumPy	Data loading, cleaning, feature engineering
Matplotlib, Plotly Express	Visualisation (static and interactive)

Scikit-learn	Linear Regression, Random Forest (Week 3)
Statsmodels — ExponentialSmoothing	ETS(A,Ad,N) Holt Damped Trend forecasting (Week 4)
Streamlit	Interactive dashboard (Week 6 stretch goal only)
GitHub	Version control — commit at end of every week

2. Weekly Breakdown

Week 1: Data Acquisition, Exploration and Understanding

Learning Objective: Understand the structure and content of GHG datasets and produce a clean, profiled dataset ready for analysis.

1.1 Data Loading

- Download the OWID CO₂ dataset from GitHub (owid/co2-data) as a CSV file
- Load into a Pandas DataFrame and display the first 10 rows, column names, data types, and shape
- Write a markdown cell explaining what each key column represents: co2, co2_per_capita, methane, nitrous_oxide, total_ghg, year, country

1.2 Data Profiling

- Report the number of null values per column as a percentage of total rows
- Identify which countries and years have the most complete data coverage
- Filter dataset to retain only rows where year $\geq$ 1990 and country is a sovereign nation (exclude aggregates like World, Asia, Europe)
- Document filtering decisions in a markdown cell with justification

1.3 Exploratory Data Analysis (EDA)

- Plot a line chart of global CO₂ emissions from 1990 to the latest year available
- Plot a multi-line chart comparing CO₂ emission trends for the top 5 emitting countries: China, USA, India, Russia, Japan
- Plot a stacked bar or area chart showing share of total global GHG by gas type (CO₂, CH₄, N₂O) per decade: 1990s, 2000s, 2010s, 2020s
- Write a 3–5 sentence summary of key patterns observed in each chart

1.4 Notebook Standards (apply from Week 1 onward)

- Every code cell must be preceded by a markdown cell explaining what the code does and why
- All charts must have titles, axis labels, and legends
- All variable names must be descriptive (no single-letter names except loop counters)
- Commit notebook to shared GitHub repository at end of every week

Week Checkpoint (Mentor Session): Loaded and profiled dataset demonstrated in notebook · Three completed EDA charts with written observations · GitHub repository set up with first commit

Week 2: Feature Engineering

Learning Objective: Transform raw emissions data into a structured, model-ready feature set that captures temporal patterns and relationships between variables.

2.1 Time-Based Features

- Add a decade column derived from the year column (e.g. 1990 → 1990s)
- Add a years_since_1990 column as a simple numeric time index for regression modelling
- For each country, compute a co2_5yr_rolling_mean column using a 5-year rolling average on annual CO₂ values

2.2 Lag Features

- For each country, create co2_lag1 (previous year CO₂), co2_lag2, and co2_lag3 columns
- Write a markdown cell explaining what lag features are and why they are useful for time-series prediction

2.3 Per-Capita and Intensity Features

- Verify co2_per_capita is correctly computed by cross-checking against co2 and population for at least 3 countries and 3 years
- Create a ghg_intensity column defined as total_ghg / gdp where both columns are available
- Note countries and years where ghg_intensity cannot be computed due to missing GDP data

2.4 Growth Rate Features

- Compute annual co2_yoy_change (year-on-year absolute change) and co2_yoy_pct_change for each country
- Identify and list the top 5 countries with the highest average annual CO₂ growth rate since 1990
- Identify and list the top 5 countries with the largest CO₂ reductions since 1990

2.5 Final Feature Dataset

Produce a clean modelling DataFrame for the 10 project countries. Required columns:

country · year · co2 · co2_per_capita · co2_5yr_rolling_mean · co2_lag1 · co2_lag2 · co2_lag3 · co2_yoy_pct_change · ghg_intensity (where available)

- Save as ghg_features.csv and commit to GitHub

Week Checkpoint (Mentor Session): Demonstrate the feature DataFrame with all engineered columns · Walk through the rolling mean and lag feature logic · Show the top 5 growth and top 5 reduction countries

Week 3: Baseline ML Models – Regression

Learning Objective: Train, evaluate, and compare supervised regression models to predict future CO₂ emissions; understand model evaluation metrics.

3.1 Problem Framing

- Write a markdown cell clearly stating the prediction task: Given features X for country C in year Y, predict CO₂ emissions for year Y+1
- Identify the target variable (co2) and input features from the Week 2 feature set
- Note on training strategy: models in this week use two different training approaches — Linear Regression is trained per country (works reliably with ~26 rows), while Random Forest is trained on the pooled dataset of all 10 countries (~260 rows) to avoid overfitting from small per-country sample sizes; both are evaluated per country for direct comparison
- Explain the choice of a supervised regression approach in a markdown cell

3.2 Train-Test Split

- For each of the 10 countries, use years 1990–2018 for training and 2019 onward for testing

The 2019–2023 test window is deliberately chosen to include the COVID-19 pandemic emissions dip (2020) and subsequent recovery — a useful real-world stress test for model robustness.

- Do NOT use random splitting — explain in a markdown cell why temporal splitting is essential for time-series data
- Report the number of training and test samples per country

3.3 Naive Baseline Model

- Implement a naive baseline: predict next year CO₂ = current year CO₂ (no-change model)
- Compute MAE and RMSE for the baseline on the test set for each country
- Plot actual vs predicted values for the baseline for 3 countries

3.4 Linear Regression

- Train a Linear Regression model using scikit-learn on the training set for each of the 10 countries
- Compute MAE and RMSE on the test set
- Plot regression line alongside actual test values for 3 countries
- Print model coefficients and write a markdown cell interpreting which features most influence predictions

3.5 Random Forest Regressor

- Important — Pooled Training Approach: With only ~26 rows per country (years 1990–2018), training Random Forest per country risks severe overfitting and produces unreliable feature importance scores. Instead, train a single Random Forest Regressor (`n_estimators=100`, `random_state=42`) on the pooled dataset of all 10 countries combined (~260 training rows). Add a `country_encoded` feature (using `LabelEncoder`) so the model can distinguish between countries of different emissions scales
- Evaluate the pooled RF model per country: compute MAE and RMSE on each country's test set (2019 onward) using the `country_encoded` feature for that country, keeping results directly comparable to the per-country Linear Regression evaluation
- Plot feature importance from the pooled RF model as a horizontal bar chart (one chart for the single pooled model, not per country); write a markdown cell interpreting which features drive cross-country emissions predictions
- Mandatory limitations markdown cell: include a notebook cell immediately before the RF training code that explains (1) why ~26 rows per country is insufficient for per-country RF (overfitting, unstable bootstrap samples, unreliable feature importance), (2) what pooling achieves and its trade-offs (learns cross-country patterns but cannot capture purely country-specific dynamics), and (3) the key teaching point that model complexity must match data availability — simple models trained on small data often outperform complex models that lack sufficient training examples

3.6 Model Comparison Table

- Produce a results table with columns: Country · Baseline MAE · LR MAE · RF MAE · Baseline RMSE · LR RMSE · RF RMSE · Best Model — note that LR MAE and LR RMSE reflect per-country Linear Regression models, while RF MAE and RF RMSE reflect the single pooled Random Forest model evaluated per country
- Write a 3–5 sentence conclusion interpreting the results

Week Checkpoint (Mentor Session): Demonstrate train-test split logic and explain why temporal splitting was used · Walk through the model comparison table and interpret at least one country result · Show the feature importance chart

Week 4: Time-Series Forecasting with ETS(A,Ad,N) — Holt's Damped Trend

Learning Objective: Apply ETS(A,Ad,N) to generate multi-year emissions forecasts with confidence intervals; understand why a damped trend model is well-suited to long-range annual emissions data.

v3 change: ARIMA replaced by ETS(A,Ad,N). Rationale: the damping parameter prevents unbounded trend extrapolation over a 20-year horizon, which is more realistic for emissions data — particularly for countries with documented slowdowns (UK, Germany). The implementation is also simpler, requiring no stationarity testing or order selection.

4.1 Concept Introduction

Write a markdown cell explaining the ETS (Error, Trend, Seasonality) state space framework. Cover:

- E (Error): additive — the model's residuals are added to the state
- T (Trend): additive damped — the trend decays toward zero over the forecast horizon via a damping parameter ϕ ($0 < \phi < 1$)
- S (Seasonality): none — annual data has no within-year seasonal cycle

Explain why ETS(A,Ad,N) is appropriate for annual emissions data:

- No within-year seasonality to model
- Damped trend prevents unbounded long-range projections (unlike a unit-root ARIMA)
- Works reliably with approximately 30 data points — fewer free parameters than many alternatives
- Physically sensible: emissions trajectories in real economies tend to slow, plateau, or gradually reverse — not grow at a constant rate indefinitely

4.2 Model Fitting

For each of the 10 countries, fit the model on the 1990–2018 training series:

```
from statsmodels.tsa.holtwinters import ExponentialSmoothing

model = ExponentialSmoothing(
    train_co2,
    trend='add',
    damped_trend=True,
    seasonal=None
)
fit = model.fit(optimized=True)
```

Print the fitted smoothing level (α), smoothing trend (β^*), and damping (ϕ) parameters for at least 3 countries. Write a markdown cell interpreting what a high vs low ϕ value implies for that country's emissions trajectory.

4.3 Forecasting to 2043

- Generate out-of-sample forecasts from 2024 to 2043 (20 years beyond the test period) for each of the 10 countries
- Include 95% confidence intervals using `fit.forecast()` or `fit.get_forecast()`
- Produce a forecast plot per country showing: historical actuals (1990–2018), fitted values, 2019–2023 holdout actuals overlaid, out-of-sample forecast to 2043 with CI shading
- Use distinct colours for actuals, fitted, holdout, and forecast

4.4 Trend Interpretation

- For at least 3 countries, write a markdown cell interpreting the forecast trend
- Discuss whether the damped projection aligns with known real-world context (e.g. UK carbon legislation, India's growth trajectory, China's peak-emissions target)
- Comment on whether the CI widens significantly over the 20-year horizon and what that implies

4.5 Forecast Summary Table

Produce a summary table with columns:

Country · 2030 Forecast (MtCO₂) · 2035 Forecast · 2040 Forecast · 2020 Actual · % Change 2020 – 2040

4.6 Model Validation

- Compute MAE and RMSE of ETS forecasts against the 2019–2023 holdout values for each country

- Add ETS MAE and ETS RMSE to the model comparison table from Week 3, creating a consolidated four-model table:

Naive Baseline · Linear Regression · Random Forest · ETS(A,Ad,N)

- Write a 3–5 sentence conclusion comparing model performance across the four approaches

Week Checkpoint (Mentor Session): Show fitted α , β^* , ϕ parameters for 3 countries and interpret the damping values · Show forecast plots for at least 3 countries with holdout overlay · Walk through the forecast summary table · Present the consolidated four-model comparison table

Week 5: Scenario Analysis (Optional — Complete Only if Time Permits)

Proceed to Week 5 only if Weeks 3 and 4 are fully complete, documented, and committed to GitHub. If time is limited, skip directly to Week 6 to finalise the notebook and prepare for the presentation.

Learning Objective: Build a what-if scenario module to simulate the emissions impact of policy interventions; develop skills in parameterised analysis and result interpretation.

5.1 Scenario Design

- Scenario A – Business as Usual (BAU): No policy change; use the ETS(A,Ad,N) baseline forecast from Week 4
- Scenario B – Moderate Mitigation: Apply a linear annual reduction rate of 2% per year to the BAU forecast starting from 2025
- Scenario C – Aggressive Mitigation: Apply a linear annual reduction rate of 5% per year to the BAU forecast starting from 2025

Write a markdown cell explaining the basis and limitations of each scenario; note these are illustrative, not scientifically calibrated.

5.2 Scenario Calculation

- For each of the 10 countries and each scenario, compute projected annual CO₂ values from 2025 to 2040
- Store results in a tidy DataFrame with columns: country · year · scenario · co2_projected
- Save as scenario_projections.csv and commit to GitHub

5.3 Scenario Visualisations

- For each of the 10 countries, produce a single line chart overlaying all 3 scenarios from 2020 to 2040 with historical actuals from 1990 to 2024 shown as a grey reference line
- Use distinct colours: blue for BAU, orange for Moderate, green for Aggressive
- Add a horizontal reference line indicating the country's 1990 emissions level as a policy benchmark
- Produce one global aggregate chart showing the sum of all 10 countries' projections under each scenario

5.4 Impact Summary

- For each country and scenario, compute total cumulative CO₂ emissions from 2025 to 2040
- Produce a grouped bar chart comparing cumulative emissions per country across the 3 scenarios

- Write a 3–5 sentence interpretation: which countries benefit most from aggressive mitigation?

Week Checkpoint (Mentor Session): Walk through the scenario DataFrame and explain the calculation logic · Show the per-country overlay charts and global aggregate chart · Present the cumulative emissions grouped bar chart with written interpretation

Week 6: Notebook Finalisation and Optional Streamlit Dashboard

Learning Objective: Finalise the notebook to professional documentation standards; optionally assemble analytical outputs into an interactive Streamlit dashboard.

6.1 Notebook Finalisation

- Ensure every section from Weeks 1–4 (and Week 5 if completed) has a clear markdown introduction and written summary conclusion
- Add a Table of Contents cell at the top of the notebook with links to each section
- Ensure all charts have consistent colour schemes, font sizes, and labelling style throughout
- Remove all debugging print statements and dead code cells

6.2 Streamlit App (Stretch Goal)

Attempt only if Weeks 3–4 are complete and time remains. Notebook quality takes priority.

- Overview section: title, project description, headline KPIs (total global CO₂ latest year, % change since 1990, number of countries analysed)
- Historical Trends: multi-line chart for user-selected countries; stacked area chart of GHG by gas type
- Country Profile: select a country to show emissions trend, per-capita trend, YoY change chart, key stats table
- Forecasts: select a country to show ETS(A,Ad,N) forecast chart to 2040 with CI and forecast summary table
- Scenario Comparison (if Week 5 complete): overlay chart of 3 scenarios and cumulative emissions bar chart
- About: data sources, methodology summary, team names, internship attribution

6.3 Interactivity Requirements (if building Streamlit app)

- At minimum: one st.selectbox for country selection, one st.multiselect for gas type, one st.radio for scenario selection
- All charts must use Plotly Express (not static Matplotlib) so they are interactive within Streamlit
- App must run without errors with: streamlit run app.py on a fresh environment with dependencies from requirements.txt

6.4 GitHub Repository Requirements (by end of Week 6)

- notebook/ghg_analysis.ipynb — the complete Jupyter Notebook

- app.py — the Streamlit application (if built)
- data/ — folder containing downloaded CSV datasets
- requirements.txt — all Python dependencies with version numbers
- README.md — project description, how to run, data sources, and team members

Week Checkpoint (Mentor Session): Walk through the finalised notebook — all sections, charts, and written summaries · Live demo of Streamlit app if completed · Review GitHub repository structure and README

Week 7: Final Presentation

Learning Objective: Deliver a structured presentation to the mentor demonstrating the full analytical workflow and consolidated findings.

The project report template and formal submission requirements will be provided by the course administration separately.

Presentation Format

- Duration: 1 hour
- Format: Slides with live notebook walkthrough

Content to cover:

- Project objective and dataset overview
- EDA key findings (Week 1)
- Feature engineering decisions (Week 2)
- Model comparison results — Naive Baseline · Linear Regression · Random Forest · ETS(A,Ad,N) (Weeks 3–4)
- ETS forecast plots and summary table (Week 4)
- Scenario analysis findings (Week 5, if completed)
- Key takeaways and limitations

Final Submission Checklist

- Jupyter Notebook: fully documented, all cells run cleanly from top to bottom
- Streamlit app (if built): runs without errors, requirements.txt complete
- GitHub repository: all files committed, README complete, repository link shared with mentor before the session
- Presentation slides: shared with mentor before the session
- Project report: to be submitted in IDEAS TIH template format (template provided by course administration)

3. Pre-Read Resource List

Tier 1 – Must Read Before Starting

- Kaggle: Intro to Machine Learning — kaggle.com/learn/intro-to-machine-learning (~3 hrs, free, runs in browser)
- Dataquest: Pandas Time Series Tutorial — dataquest.io/blog/tutorial-time-series-analysis-with-pandas (~1.5 hrs)
- Scikit-learn Beginner Tutorial — scikit-learn.org/stable/tutorial/basic/tutorial.html (~1 hr)
- Statsmodels Exponential Smoothing docs — statsmodels.org/stable/tsa.html#exponential-smoothing (use in Week 4)

Tier 2 – Pick Up During Project as Needed

- Hyndman & Athanasopoulos, Forecasting: Principles and Practice Ch 7–8 (ETS models) — otexts.com/fpp3/ets.html
- Plotly Express documentation — plotly.com/python/plotly-express (use in Week 6)
- Streamlit Get Started tutorial — docs.streamlit.io/get-started (use in Week 6, if attempting Streamlit)
- W3Schools Python Machine Learning — w3schools.com/python/python_ml_getting_started.asp (quick reference)

Domain Reference

- Our World in Data: CO₂ and GHG Emissions — ourworldindata.org/co2-and-greenhouse-gas-emissions
- Machine Learning for Climate Change — Rolnick et al. (2022), [arXiv:1906.05433](https://arxiv.org/abs/1906.05433)
- IPCC AR6 Summary for Policymakers — ipcc.ch/report/ar6/wg1

4. Version History

Version	Date	Change
v1	Jun 2026	Initial scope document
v2	Jun 2026	Train/test split rationale added; Week 5 marked optional; Streamlit marked stretch goal
v3	Jun 2026	Week 4 forecasting model changed from ARIMA(1,1,1) to ETS(A,Ad,N) Holt Damped Trend. Rationale: damped trend prevents unbounded long-range extrapolation and better captures emissions slowdowns. Pre-read updated to FPP3 Ch 7–8 (ETS) in place of ARIMA tutorial.
v4	Jun 2026	Week 3 Random Forest training strategy changed from per-country to pooled (all 10 countries, ~260 rows). Rationale: ~26 rows per country is insufficient for reliable RF;

		pooling provides adequate training data. country_encoded added as RF feature. Mandatory limitations markdown cell added to Section 3.5. Model comparison table updated to note LR is per-country and RF is pooled.
--	--	--